

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

Inputs	Process	Outputs
Quantity of notebooks purchased Quantity of pens purchased Amount of money paid	Multiply the number of notebooks by ₱45 to find the notebook expense. Multiply the number of pens by ₱12 to find the pen expense. Add both expenses to determine the overall cost. Subtract the total cost from the payment to determine the change.	Total amount to be paid Amount of change received

DECOMPOSITION

- Get the inputs
- Calculate notebook cost
- Calculate pen cost
- Calculate total cost
- Calculate change
- Display the results

PSEUDOCODE

START

READ number of notebooks

READ number of pens

READ payment amount

notebookExpense ← number of notebooks × 45
penExpense ← number of pens × 12

total ← notebookExpense + penExpense
change ← payment amount - total

DISPLAY notebookExpense
 DISPLAY penExpense
 DISPLAY total
 DISPLAY change

END

DRY-RUN CALCULATION:QUICK ACTIVITY ONE

CASE TEST 1:

- Notebook: 3
- Pens: 2
- Payment: ₱200

STEP	VARIABLE	CALCULATION	RESULT
1	Notebookcost	3 x ₱45	₱135
2	Pencost	2 x ₱12	₱24
3	totalcost	₱135 + ₱24	₱ 159
4	change	₱200 - ₱159	₱41

CASE TEST 2:

Input:

- Notebook: 2
- Pens: 3
- Payment ₱ 150

STEP	VARIABLES	CALCULATION	RESULT
1	notebookcost	2 x ₱45	₱90
2	pencost	3 x ₱12	₱36
3	totalcost	₱90 + ₱36	₱126
4	change	₱150 - ₱126	₱24

